

StatMind

Process Statistics Report

Etch_Rate_nm_min

Generated: 2026-05-13 06:37 | Process: RIE Etch — Chamber A | n = 80

Executive Summary

Normality: Normal — Standard Cp/Cpk (parametric)

Capability: Cpk = **1.041** — Marginal. Sigma level: 3.124. PPM (within): 1,783

SPC: I-MR chart — **39 rule violation(s)** Out of Control — 39 alarms, immediate investigation required

GRR: %GRR = **29.6%** — Marginal. ndc = 4

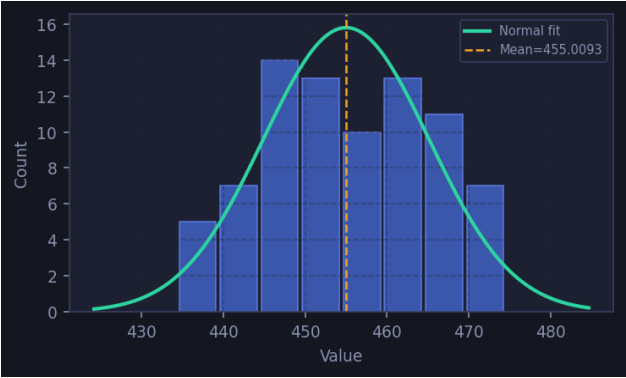
CAPA: **Critical** — Litho CD — Low Cpk (Dose or Focus Drift). Disposition: **Hold**

1. Normality Analysis

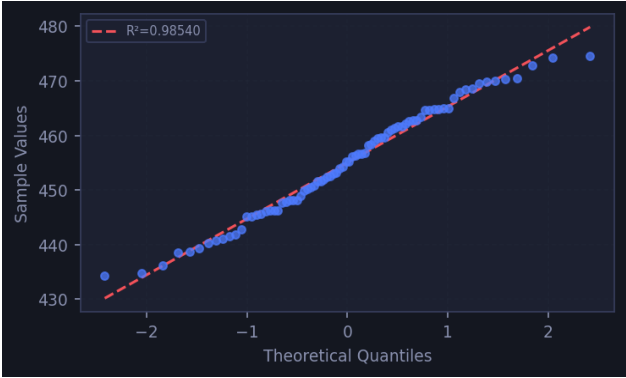
Verdict: Normal (Confidence: High)

Statistic	Value	Statistic	Value
n	80	Skewness	-0.0614
Mean	455.00928	Kurtosis	2.1217
Std Dev	10.21494	Min	434.32264

Test	Statistic	p-value	Result
Shapiro-Wilk	0.97903	0.21249	Normal
Anderson-Darling	0.39690	0.36860	Normal
Ryan-Joiner	0.99267	0.99990	Normal



Histogram with Normal Curve Overlay



Normal Probability Plot (R²=0.98540)

- Cp/Cpk and Pp/Ppk calculations are appropriate
- Xbar-R or Xbar-S control charts are appropriate
- Standard confidence intervals on Cpk are valid

2. Process Capability

Verdict: Marginal — Sigma Level: 3.124σ

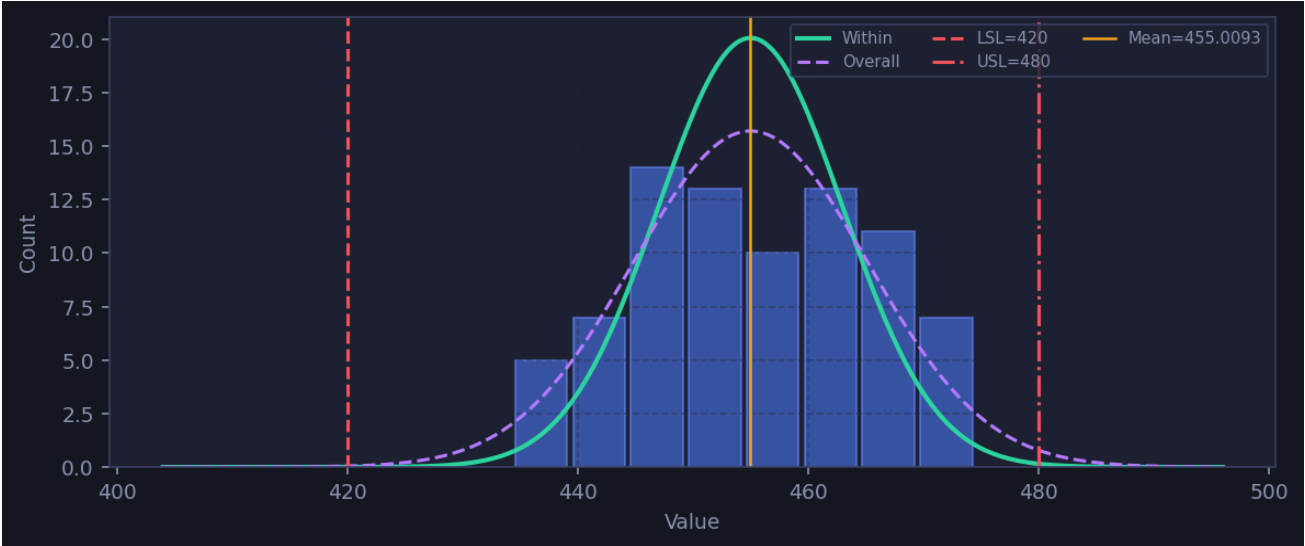
Cpk=1.041 between 1.00–1.33 — Barely capable. Improvement recommended.

Index	Value	Index	Value	Index	Value
Cp	1.2501	Cpk	1.0414	Cpm	1.0595
Pp	0.9790	Ppk	0.8155	CPU/CPL	1.0414 / 1.4588

Parameter	Value	Parameter	Value
USL	480	LSL	420
Target	450.0000	Mean	455.00928
σ Within	7.99933	σ Overall	10.21494

PPM Within	1,783.5	PPM Overall	14,425.5
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Confidence	Lower	Cpk	Upper
90%	0.8919	1.0414	1.1908
95%	0.8633	1.0414	1.2194
99%	0.8074	1.0414	1.2754



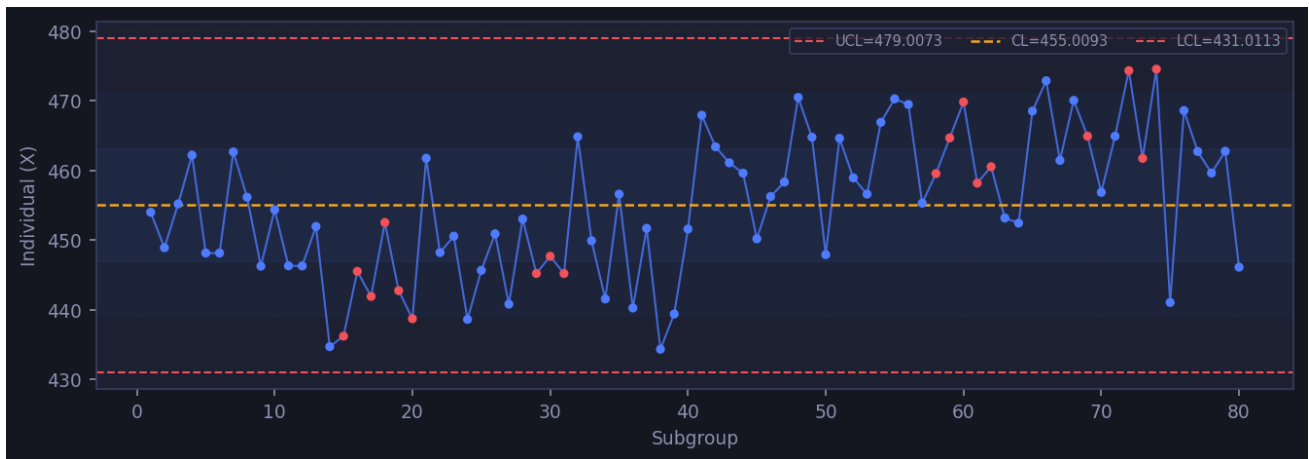
Capability Histogram with Spec Limits, Within and Overall Normal Curves

- Expected 1,783 PPM defects — investigate assignable causes and tighten process control.
- Process mean is dangerously close to USL. Consider recipe adjustment or tighter APC setpoint.

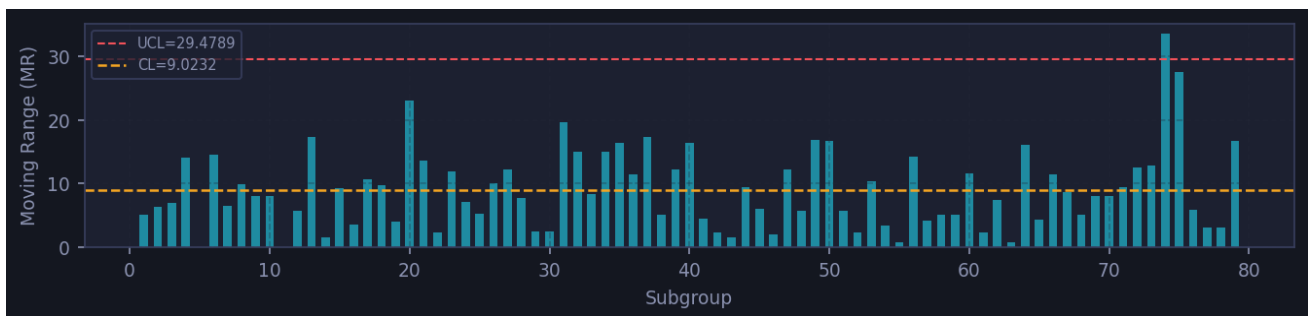
3. SPC Control Charts

I-MR Chart — Out of Control — 39 alarms, immediate investigation required

Parameter	Value	Parameter	Value
UCL	479.00726	LCL	431.01130
Center Line	455.00928	Process Sigma	7.99933
n Points	80	Total Alarms	39



Individual (X) Chart — alarm points highlighted in red



Moving Range (MR) Chart — alarm points highlighted in red

Point	Rule	Value	Description
15	WE2	436.2007	2 of 3 consecutive points beyond -2σ
15	WE3	436.2007	4 of 5 consecutive points beyond -1σ
16	WE2	445.5017	2 of 3 consecutive points beyond -2σ
16	WE3	445.5017	4 of 5 consecutive points beyond -1σ
16	WE4	445.5017	8 consecutive points below centerline
17	WE3	441.8974	4 of 5 consecutive points beyond -1σ
17	WE4	441.8974	8 consecutive points below centerline
17	NE2	441.8974	9 consecutive points below centerline (shift)
18	WE3	452.5140	4 of 5 consecutive points beyond -1σ
18	WE4	452.5140	8 consecutive points below centerline
18	NE2	452.5140	9 consecutive points below centerline (shift)
19	WE3	442.7358	4 of 5 consecutive points beyond -1σ
19	WE4	442.7358	8 consecutive points below centerline
19	NE2	442.7358	9 consecutive points below centerline (shift)
20	WE3	438.7016	4 of 5 consecutive points beyond -1σ
20	WE4	438.7016	8 consecutive points below centerline

20	NE2	438.7016	9 consecutive points below centerline (shift)
29	WE4	445.1949	8 consecutive points below centerline
30	WE4	447.6665	8 consecutive points below centerline
30	NE2	447.6665	9 consecutive points below centerline (shift)
...			+ 19 more violations

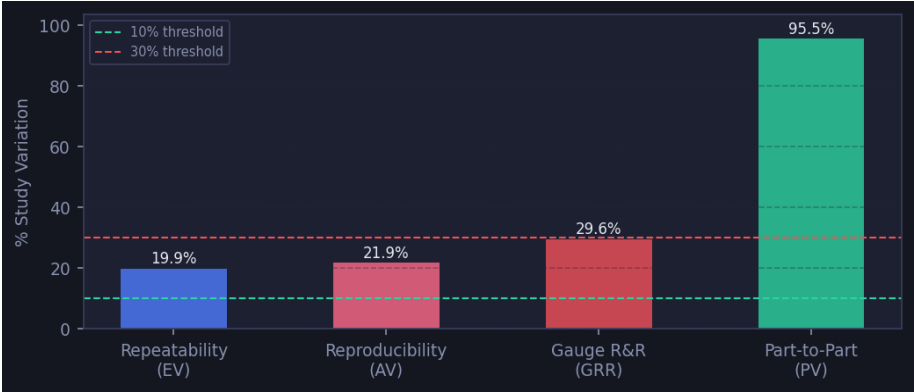
- NE2: 12 occurrence(s)
- WE2: 3 occurrence(s)
- WE3: 8 occurrence(s)
- WE4: 16 occurrence(s)

4. Gauge R&R; (MSA)

Verdict: Marginal — %GRR = 29.6% between 10–30% — Conditionally acceptable. Improve if possible.

Study Design	Value	Key Metric	Value
Method	ANOVA	ndc	4
Parts	10	%GRR	29.6%
Operators	3	% Tolerance	31.37%
Replicates	2	Total Readings	60

Source	Variance	Std Dev	% Contribution	Study Var	% Study Var
Repeatability (EV)	4.45e+00	2.110497	4.0%	12.66298	19.9%
Reproducibility (AV)	5.39e+00	2.321288	4.8%	13.92773	21.9%
Operator × Part	5.44e-01	0.737627	0.5%	4.42576	7.0%
Gauge R&R; (GRR)	9.84e+00	3.137288	8.7%	18.82373	29.6%
Part-to-Part (PV)	1.03e+02	10.139724	91.3%	60.83834	95.5%
Total Variation (TV)	1.13e+02	10.613980	100.0%	63.68388	100.0%



GRR Component Breakdown (% Study Variation)

Source	SS	df	MS	F	p-value
Parts	5601.837615	9	622.426402	139.739	0.0000
Operators	204.856170	2	102.428085	22.996	0.0000
Parts × Operators	99.762900	18	5.542383	1.244	0.2903
Repeatability	133.625881	30	4.454196	—	—
Total	6040.082566	59	0.000000	—	—

- Reproducibility (21.9%) dominates — operator influence. Standardize measurement procedure and provide operator training.
- ndc = 4 — gauge can make only 4 distinct categories. Industry standard requires $\text{ndc} \geq 5$.
- GRR consumes 31.4% of tolerance — measurement uncertainty is too high for this specification.

5. CAPA Report

Severity: Critical | Confidence: High | Rule: LITHO-001 | Process: Lithography

Litho CD — Low Cpk (Dose or Focus Drift)

Lithography CD capability is below threshold. Process mean offset indicates dose or focus drift.

- $\text{Cpk} = 1.041 < 1.33$ (below industry standard)
- $\text{Cp} = 1.250$ vs $\text{Cpk} = 1.041$: centering gap of 0.209 indicates off-center process
- Expected 1,783 PPM defects (within-subgroup estimate)
- 39 SPC rule violation(s) detected: WE2, NE2, WE3, WE4
- Gauge R&R; = 29.6% (Marginal — 10–30%)
- $\text{ndc} = 4$ (inadequate — <5 required)

Confidence: High

Dose or focus offset in scanner causing systematic CD bias across wafer or lot.

In lithography, CD is primarily controlled by dose (energy) and focus (z-height). Centering failure ($\text{Cp} \gg \text{Cpk}$) means the scanner dose or focus setpoint has drifted. APC system may not be correcting fast enough or the correction model has degraded.

Alternative causes:

- Resist coating thickness variation altering effective dose sensitivity
- BARC (bottom anti-reflective coating) thickness drift affecting CD
- Reticle CD error or particle contamination on reticle
- Scanner illumination uniformity degradation

P1	Pull focus-exposure matrix (FEM) data from last qualification run. Determine current best focus and best dose. Compare to recipe setpoints — update if $\Delta > \pm 2\text{nm}$ focus or $\pm 0.3\%$ dose. Expected CD centering to within $\pm 1\text{nm}$ of target	Immediate Process Engineer
P1	Inspect reticle for particles or damage using reticle inspection tool. Clean reticle if contamination found. Rules out reticle contamination as cause	Immediate Manufacturing
P1	Check scanner APC model validity date. If >2 weeks since last calibration, run APC calibration wafers. Restores APC correction accuracy	1 week Equipment

P2	Measure BARC thickness on last 10 wafers. If $\geq \pm 5\text{\AA}$ drift from target, investigate BARC coating process. Rules out resist stack contribution	1 week Metrology
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APC	Implement APC dose and focus feedback using after-develop inspection (ADI) CD data. Correct every lot.	2 weeks Equipment
SPC	Add resist CD SPC chart at ADI. Use as leading indicator for before-etch yield risk.	2 weeks Process Engineer
PM	Schedule reticle inspection every 500 exposures or weekly, whichever comes first.	1 week Manufacturing

Disposition: Hold	Based on Litho CD — Low Cpk (Dose or Focus Drift) — Lithography CD capability is below threshold. Process mean offset indicates dose or focus drift.
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Containment: Hold all wafers exposed on affected scanner since last confirmed good lot. Inspect ADI CD on 3 wafers per lot before releasing.